AGENTIC WEARABLE EARLY WARNING

Wearable Health Agent

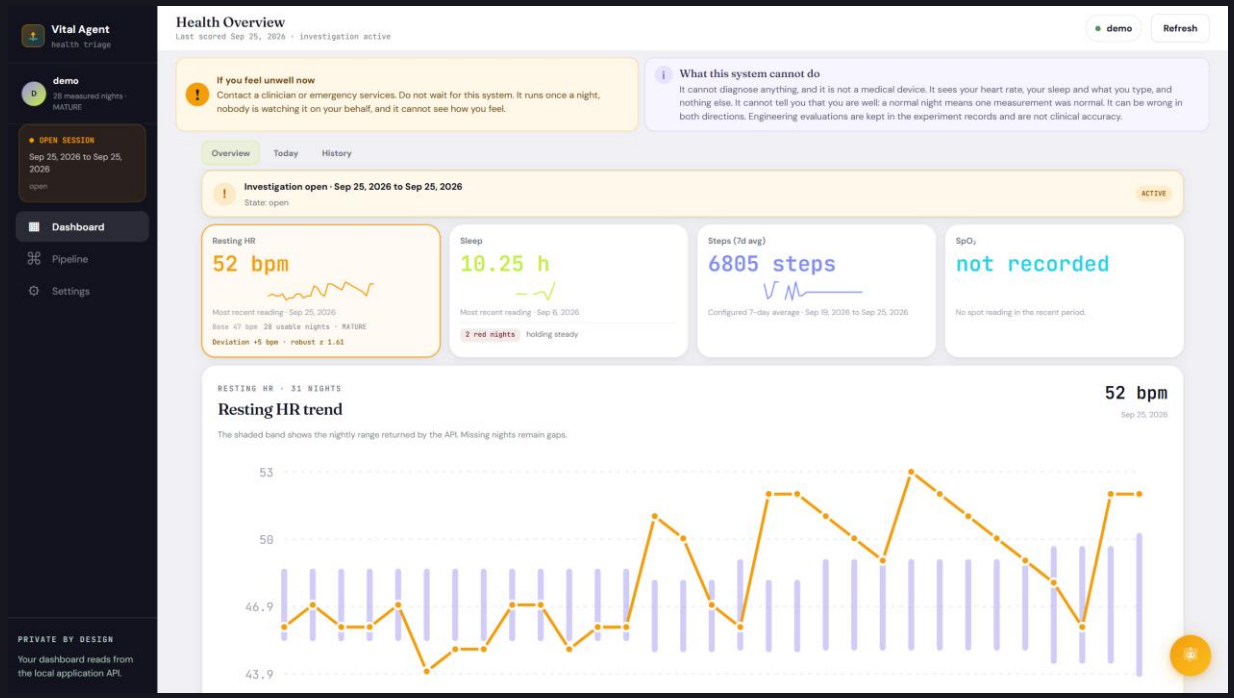
An AI agent that investigates why your sleeping heart rate changed: it gathers evidence, decides whether to ask you, and explains with sources. A math harness wakes it up.

AI AGENT: INVESTIGATES

HARNESS: WAKES IT UP

SAFETY: FIXED RULES

Nathaneal Brian · Dhruv Sethi
Harshini Balasubramanian · Tham Ji Yee



The new dashboard · Health Overview

Your watch can tell that something changed. Not why.

A night 4 bpm above your normal looks the same to a wrist sensor whatever caused it.

Possible cause	Present on any night	Typical rise
Luteal phase of the cycle	45.0 %	+3.91 bpm
Acute stress	35.0 %	+1.50 bpm
Short or broken sleep	34.8 %	+1.00 bpm
Alcohol	29.9 %	+2.60 bpm
Unrecorded exertion	8.0 %	+2.91 bpm
Infection	3.8 %	+1.60 bpm
… plus vaccine, travel, medication, recovery, sensor artefact, none		

12 causes in data/knowledge/causes.yaml; each figure carries its own citation there.

The cause we want is rare

Infection is present on 3.8 % of nights, less often than alcohol, stress or the cycle.

So the system asks

The sensor can't tell these apart. A person can, with one well-chosen question.

What this system can and cannot do

The system is built to notice sustained personal change, not to name a disease.

What it can do

1. **Notice**
   Compare sleeping heart rate with the person's own recent nights.
2. **Investigate**
   Combine the signal with context and local health information.
3. **Ask only if useful**
   Pause for a short answer when it could change the conclusion.

The AI agent can: read approved local data, retrieve health passages, ask a bounded question, and explain its reasoning.

What it cannot do

- ✕ It checks sleeping heart rate only.
- ✕ It cannot monitor daytime health, detect sleep apnoea, or diagnose.
- ✕ Normal readings do not prove someone is well.

The AI agent cannot: no open-web browsing, arbitrary writes, diagnosis, emergency decision, or lowering the safety level.

If someone feels unwell, contact a clinician or emergency services. Do not wait for the system.

Research prototype · Not a diagnosis tool or medical device

An AI agent, with a math harness and a safety floor around it

Three different kinds of machine, colour-coded the same way the new dashboard colours its trace.

HARNESS

Arithmetic, no AI

- Runs every night, automatically
- Builds your personal baseline
- Scores each night GREEN / YELLOW / RED
- Opens an investigation on 2 RED nights in a row
- Same input → same answer, every time

AI AGENT

Bounded investigator

- Wakes only when a night is flagged
- Chooses read-only tools to gather evidence
- Writes the question and the explanation
- Chats with you, citing a local library
- Can't alert, diagnose or lower urgency

SAFETY FLOOR

Fixed rules

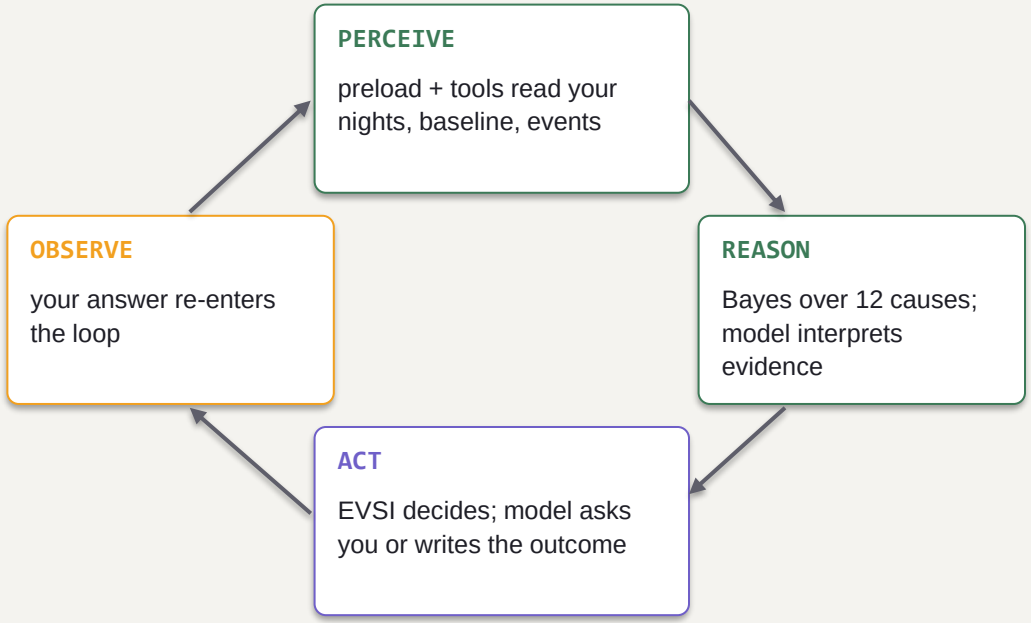
- Reads everything you type, before the AI does
- Phrase rules + extreme readings + SpO_2
- Published scoring rule for chest pain
- AI may add a concern, never remove one
- Urgency only ever goes up

An AI agent, not a chatbot

A chatbot waits for you and replies. This agent starts itself, plans, uses tools, remembers, acts, and checks its own work.

	A chatbot	This AI agent
Who starts it	you type a message	wakes itself when a night is flagged
Goal	reply to the last message	explain a change: which of 12 causes?
Plans its steps	no	picks which tools to call, in what order, loops
Uses tools	no	6 read-only tools: your data, history, library
Memory	the chat window	saved state graph; pauses for days, resumes
Acts	writes text	decides to message you on Telegram; writes outcome
Checks itself	no	critique strips unsourced claims; human review

What it is not allowed to do is just as deliberate: it can't raise an alert, choose urgency, or diagnose. Those stay with the math and the rules (previous section).



Proof from the demo log, 13 Sep

```
06:36:34  you: "I have had a fever since last night, and chills"
06:36:41  agent → retrieve_clinical_evidence("fever"), ("chills")
```

Nobody scripted those searches. It followed the new evidence.

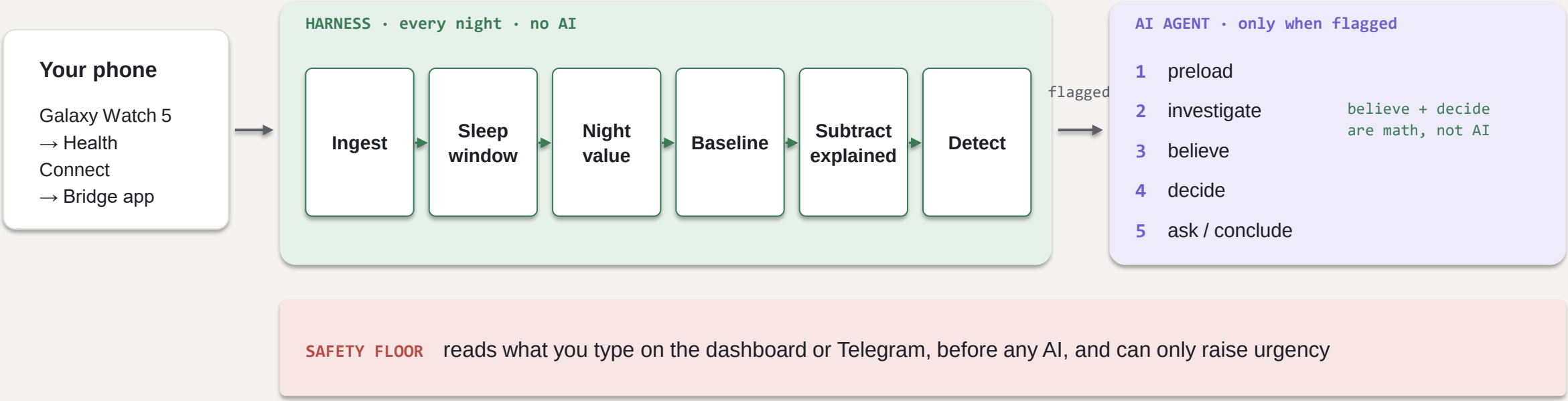
Who decides what

Every decision that affects the person is made by math or fixed rules. The AI's job is language.

Decision	Made by	How
Was last night raised?	Math (harness)	residual vs your own baseline, 3 / 4 bpm thresholds
Open an investigation?	Math (harness)	2 consecutive RED nights, ≥ 7 nights of your own data
Which cause is most likely?	Math (harness)	Bayes' rule over 4,096 cause combinations
Is a question worth interrupting you?	Math (harness)	EVSI score minus interruption cost
Which tools to look things up with	AI agent	chosen in a loop, closed list, read-only, ≤ 25 calls
The wording of the question & reply	AI agent	then checked: every clinical claim needs a source span
How urgent is this? See a doctor?	Fixed rules	phrase rules, Marburg Heart Score, SpO_2; AI can only raise

The whole journey, from wrist to answer

The harness runs every 30 minutes on new data. The agent only enters when the harness flags a night.



Dashboard

React + Vite, 127.0.0.1:8443

Telegram

questions with buttons, and chat

Audit log

hash-chained, tamper evident

Local library

1,849 health passages for RAG

Input data: what goes in, and what each input is for

One signal drives the alert. Everything else is context, a safety check, or something you tell it.

Input parameter	Source	Resolution	Used for
Heart rate	watch	~1 sample / min	THE signal: one median per night
Sleep stages	watch	per session	which minutes count (awake spans removed)
Exercise sessions	watch	start / end time	subtract a recorded evening workout
Steps, distance, calories	watch	daily buckets	context only
Blood oxygen (SpO_2)	watch	spot readings	safety rule, needs ≥ 10 overnight
What you type / tap	dashboard, Telegram	free text, buttons	safety floor + belief update
Profile: age, sex, cycle, heart condition	Settings	standing facts	chest-pain rule age gate, luteal cause

Not available from this watch: heart rate variability, skin temperature, breathing rate. A hard limit, not a missing feature.

Step 1 · Turn ~400 readings into one number per night

Measure only while you are actually asleep, then take the middle value.

1

Find your sleep

Group sleep-stage rows into sessions (split on a 60 min gap). Drop AWAKE spans. Keep the main session ending on this date; naps are not pooled.

2

Filter readings

Keep 25 ≤ HR ≤ 220 bpm. Count and record what was dropped.

3

Take the median

The middle value, so a handful of spikes can't move it the way a mean would.

4

Check coverage

Need ≥ 180 distinct minutes (3 h) inside the window, or the night is unusable.

THE FORMULA

```
W  = recorded sleep − awake spans
HR = { h in W : 25 ≤ h ≤ 220 }

RHR_night = median(HR)

usable  ⇔  coverage(W) ≥ 180 min
```

No sleep record yet?

Fall back to local midnight–07:00 and label the night 'clock'. The new dashboard says so on the night's record.

Step 2 · Your baseline: what "normal for you" means

A centre (your usual sleeping heart rate) and a scale (how much you normally wobble night to night).

BASELINE

```
n = usable nights in the last 28
w = n / (n + 7)

if n ≥ 7:
   centre = median(RHR)
else:
   centre = w·median(RHR) + (1−w)·62.75

MAD    = median |RHR − median(RHR)|
scale = max( w·1.4826·MAD + (1−w)·3.71 , 1.5 )
```

62.75 bpm: population mean (Bloomfield et al. 2024, n = 525). 3.71 bpm: median within-person spread, measured by this project on 107 people's archived wearable data. Both stop mattering as your own nights arrive.

Nights	Rung	Can flag?
0–2	COLD	no
3–6	PRIOR_DOMINATED	no
7–13	SHRINKING	yes
14–27	PERSONAL	yes
28+	MATURE	yes

Why 7 nights?

NightSignal (Alavi et al. 2022, Nature Medicine): for over 80 % of people the baseline settled within seven nights.

Step 3 · How an alert is triggered (math, not AI)

Subtract only what we can verify, colour the night, and require it to persist.

NIGHTLY SCORE

```
deviation = RHR_night − centre
explained = recorded workout effect
residual  = deviation − explained
robust_z  = residual / scale
```

Workout ended before sleep	Subtracted
0–2 h	2.91 bpm
2–4 h	1.87 bpm
4–6 h	1.58 bpm
> 6 h or none	0

GREEN

residual < 3 bpm

YELLOW

3 ≤ residual < 4 bpm

RED

residual ≥ 4 bpm

OPEN AN INVESTIGATION ⇔

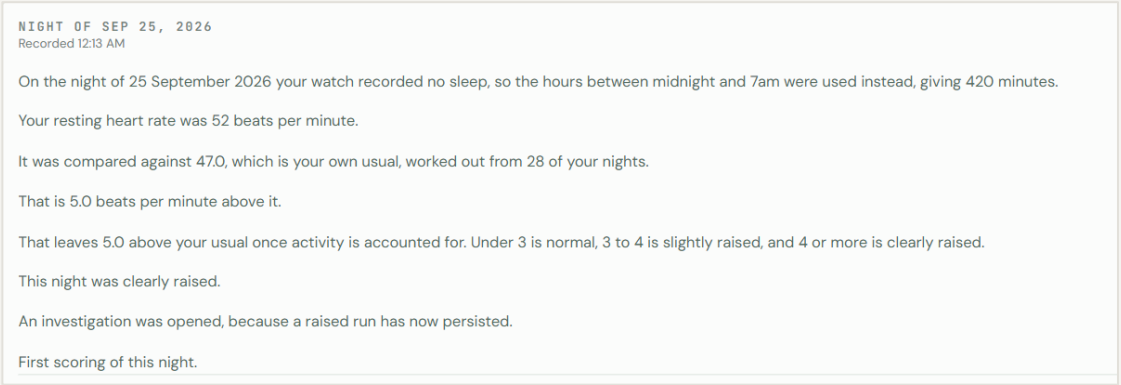
- ✓ tonight is RED
- ✓ last night was also RED
- ✓ ≥ 7 usable nights of your own
- ✓ the change is a rise, not a fall

Thresholds from NightSignal (Alavi et al. 2022), measured on 3,318 people.

Worked example, straight from the demo's record

The Pipeline → Everything the system did tab writes the same arithmetic out as sentences.

	24 Sep	25 Sep
RHR_night	52 bpm	52 bpm
centre (28 nights, MATURE)	47.0	47.0
deviation	+5.0	+5.0
explained (no workout)	0.0	0.0
residual	+5.0	+5.0
level	RED	RED
flagged?	no (first)	YES → open


NIGHT OF SEP 25, 2026
Recorded 12:13 AM

On the night of 25 September 2026 your watch recorded no sleep, so the hours between midnight and 7am were used instead, giving 420 minutes.

Your resting heart rate was 52 beats per minute.

It was compared against 47.0, which is your own usual, worked out from 28 of your nights.

That is 5.0 beats per minute above it.

That leaves 5.0 above your usual once activity is accounted for. Under 3 is normal, 3 to 4 is slightly raised, and 4 or more is clearly raised.

This night was clearly raised.

An investigation was opened, because a raised run has now persisted.

First scoring of this night.

`robust_z = 5.0 / 3.1 ≈ 1.61`

Well below the extreme-heart-rate safety rule (|z| > 6), so this goes to the agent, not straight to urgent care.

The safety floor runs first, and the AI can only make it louder

Fixed rules read your words before any model does. The model's flags are merged by set union.

A — Urgent phrases

chest pain, fainted, confusion, can't breathe, bleeding, stiff neck + fever

B — Symptoms over time

fever, aches, vomiting, exhaustion: 2+ within 2 nights, or one that persists

C — Model's second opinion

the AI reads the same text and may ADD flags, never remove

```
flags = A ∪ B ∪ C          (code, not model)
urgency = max(over whole conversation)
```

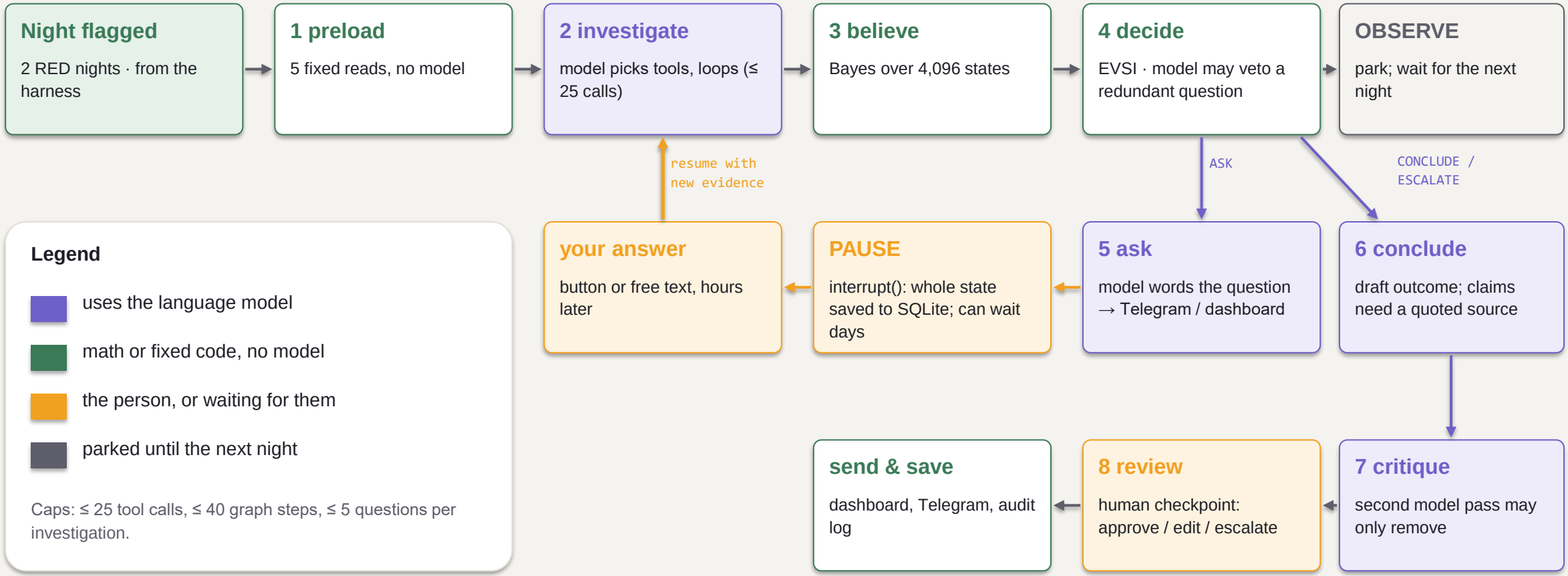
Rules that use numbers, not words

Check	Threshold	Result
Extreme heart rate	|robust_z| > 6	escalate
Overnight SpO_2 (median, ≥10 reads)	≤ 92 %	urgent
	93–94 %	same-day review
Marburg Heart Score (chest pain, age ≥ 35)	≥ 3 of 5	urgent
	0–2 of 5	see a doctor today

Marburg: Bösner et al. 2010, CMAJ, 1,199 patients, sensitivity 87.1 %, specificity 80.8 %. SpO_2 bands: NHS COVID Oximetry @home. The model asks the five questions; the arithmetic decides.

The agent graph (LangGraph): every node, every path

A state machine whose state is saved to SQLite after every node. Purple nodes use the language model.



What the agent may touch: tools and guardrails

A closed registry of read-only tools. The model chooses which to call; code enforces everything else.

Tool the model can call	Returns	Demo, 13 Sep
get_nightly_features	readings, coverage, sleep hours	
get_personal_baseline	centre, band, maturity	
get_known_events	workouts, steps, SpO_2	5–13 Sep → 8 items
get_data_quality	coverage, dropped readings	
find_similar_past_episodes	your earlier investigations	→ 0 items
retrieve_clinical_evidence	passages from the local library	3 searches, 5 each

Enforced in code, not in the prompt

Unknown tool → refused and audited · subject id replaced by the investigation's own · dates clamped to 90 days and never the future · no web access, no writes

Question gates

Baseline ≥ 7 nights
Max 2 unsolicited asks per week
Max 5 questions per investigation
Only one question waiting at a time

One exception

If you've said you feel unwell, a follow-up rule can ask even when the weekly budget is spent. A budget exists to stop pestering a well person, never to silence an unwell one.

Agent step · weigh 12 causes at once with Bayes' rule

Pure arithmetic inside the agent loop: every on/off combination of the 12 causes is scored exactly.

BELIEF UPDATE

$$s \in \{0,1\}^{12} \quad \rightarrow \quad 2^{12} = 4{,}096 \text{ combinations}$$

$$\mu_s = \Sigma_{c \text{ on in } s} \text{ effect}_c \cdot w_c(t)$$
$$\sigma^2_s = 1.2^2 + \Sigma_{c \text{ on}} (\text{sd}_c \cdot w_c(t))^2$$
$$\text{residual} \mid s \sim \text{Normal}(\mu_s, \sigma^2_s)$$

$$P(s \mid \text{evidence}) \propto P(s) \cdot P(\text{residual}|s) \cdot P(\text{answers}|s)$$
$$P(\text{cause } c) = \Sigma_{s : c \text{ on}} P(s \mid \text{evidence})$$

w_c(t) is each cause's shape over nights: alcohol is full on night 1 and gone by night 2; infection rises over ~3 nights then fades. Watch evidence (workout, short sleep, poor coverage) and your answers add log-factors.

A recorded run: two nights at +5 bpm

Cause	Before	After "feverish"
Infection	7.6 %	40.5 %
Stress	81.5 %	77.7 %
Alcohol	55.4 %	54.1 %
Short sleep	48.1 %	47.5 %
Sensor artefact	9.2 %	7.6 %

Each is the probability that cause is present. They don't add to 100 %, because several can be true at once.

EVSI: should the agent interrupt you with a question?

Expected Value of Sample Information = how much better our decision gets, on average, after hearing your answer.

1. **What would we do right now?**
   Pick the cheaper mistake: treat as possible illness, or as benign.

2. **Imagine each possible answer**
   Use Bayes to see how each answer would move p(infection).

3. **Would any answer change our choice?**
   If not, the question is worth 0. If yes, how much better on average?

4. **Is that worth interrupting you?**
   Subtract the interruption cost. Ask only if the result is > 0.

COST OF BEING WRONG

```
U(miss illness) = −9    U(false alarm) = −1
tipping point p* = 1 / (1 + 9) = 0.10
```

VALUE OF A QUESTION

```
EVSI(q) = Σ_a P(a)·best EU after a − best EU now
score(q) = EVSI(q) − cost_q × 0.15
```

p(infection)	0.02	0.05	0.08	0.10	0.20	0.30	0.40
EVSI	0.016	0.195	0.428	0.585	0.370	0.155	0.000
score	−0.134	+0.045	+0.278	+0.435	+0.220	+0.005	−0.150
ask?	no	YES	YES	YES	YES	YES	no

The quiet band. Very unlikely or already likely: no answer changes the decision, so it doesn't ask. Symptom question, cost 1.0, computed with the current code.

EVSI worked example: is the symptom question worth it at p = 8 %?

Every number below is the project's own formula, done by hand, and matches what the code returns.

```
WITHOUT ASKING  (p = 0.08)

treat as illness: −1 × 0.92 = −0.92
treat as benign:  −9 × 0.08 = −0.72  ← best
```

Answer	P(answer)	p(inf) after	Best choice	EU
Nothing unusual	0.748	0.016	benign	−0.144
Throat or nose	0.088	0.271	illness	−0.729
Feverish or chills	0.048	0.420	illness	−0.580
Aches or heavy limbs	0.062	0.258	illness	−0.742
Something else	0.054	0.148	illness	−0.852

```
EVSI = −0.292 − (−0.72) = 0.428 → score = 0.428 − 0.15 = +0.278
→ ASK
```

The real run, 13 Sep: 9 candidate questions scored

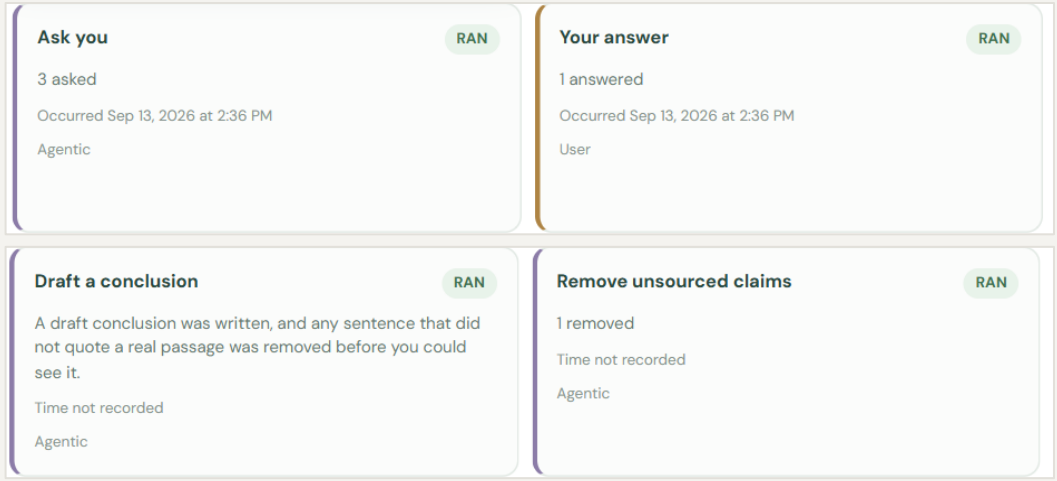
Question (probe)	Score	
symptoms	+0.333	ASKED
medication	−0.060	
travel	−0.076	
vaccination	−0.086	
exertion	−0.120	
sleep quality	−0.135	
alcohol	−0.150	
stress	−0.150	

p(infection) was 14.8 %. Symptoms EVSI 0.483 − 0.15 = +0.333. Every other question would not move the decision enough to justify interrupting, so the agent asked exactly one.

A real investigation, step by step (demo, 13–17 Sep)

What the agent actually did, read from the demo's record. Nothing on this slide is a mock-up.

12–13 Sep	harness	Two RED nights → investigation opens
06:36:07	model	calls find_similar_past_episodes, get_known_events, retrieve("elevated nocturnal heart rate")
believe	math	p(infection): 3.8 % base rate → 14.8 %
decide	math	EVSI: symptom question +0.333; 8 others negative → ASK
ask	model	writes "Have you noticed any unusual symptoms today?" → graph pauses
06:36:34	you	"I have had a fever since last night, and chills"
06:36:41	model	re-investigates: retrieve("fever"), retrieve("chills") → p(infection) 19.7 %
14, 16 Sep	model	rewords and asks again; no reply → nothing inferred from silence
critique	code	strips "Fever and chills were reported…": no source span



PIPELINE · THIS NIGHT, STEP BY STEP · 13 SEP

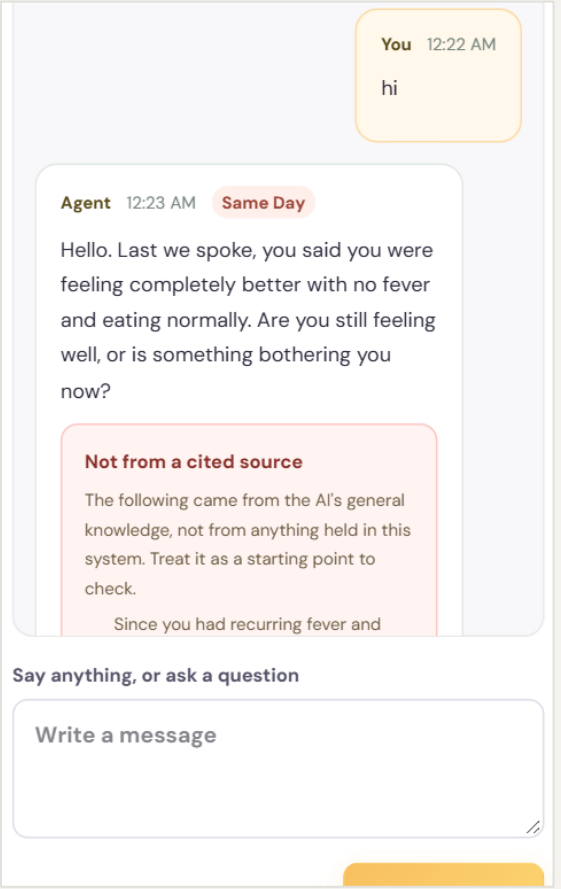
Outcome

Disposition UNRESOLVED · action: monitor · confidence: low
Escalation criteria selected from cited text, not invented: "Seek urgent medical attention… if you develop chest pain, shortness of breath at rest, fainting, or sudden confusion."

How the AI helps in chat, and what checks it

You can ask anything, on the dashboard or Telegram. Two model calls per turn, wrapped in rules.

1	**Safety floor**	fixed rules read your words first
2	**Retrieve**	top passages from the local library (RAG)
3	**Reply model**	answers in context, T = 0.3; every clinical claim carries a quoted span
4	**Grounding check**	code confirms each span is really in a passage; if not, the claim is stripped
5	**Extractor model**	separate call, T = 0, only pulls out answers a rule needs
6	**Rule decides**	urgency from the rule; it only ever rises, kept across restarts



Real chat 1: the agent asks follow-ups, the rule sets urgency

Conversations from the demo. The AI writes the reply and the follow-up question; fixed rules decide how urgent it is.

12 SEP

YOU

I feel a bit dizzy today

AGENT

Thanks for letting me know. Dizziness can have many causes. Have you felt like you might faint or lose consciousness?

Source: MedlinePlus · Balance Problems
Rule: dizziness is tracked, no escalation

14 SEP

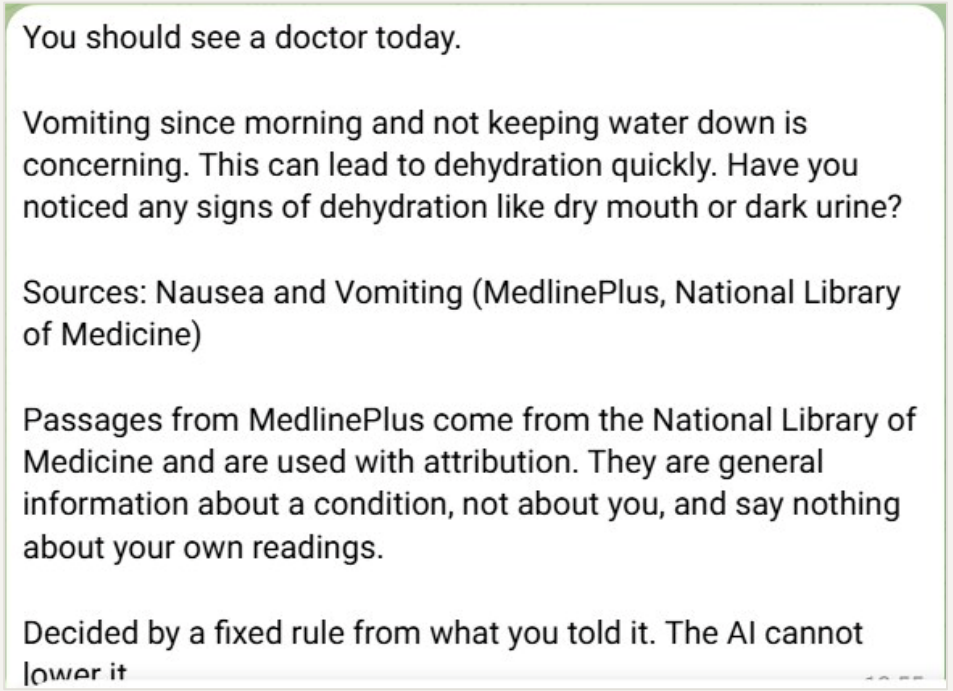
YOU

I have been vomiting since this morning and cannot keep water down

AGENT

Vomiting since morning and not keeping water down is concerning. This can lead to dehydration quickly. Have you noticed any signs of dehydration like dry mouth or dark urine?

Source: MedlinePlus · Nausea and Vomiting
Rule: fever (13 Sep) + vomiting + poor intake
→ SEE A DOCTOR TODAY



The real Telegram message. Rule verdict first, then the AI's reply and its source, then: "Decided by a fixed rule from what you told it. The AI cannot lower it."

Real chat 2: it remembers you, and urgency never silently drops

Multi-turn memory across days, with every reply checked. Text not backed by a source is labelled as such.

15 SEP

YOU

I feel a lot better today, the fever has gone

AGENT

Good to hear you're feeling better and the fever is gone. Since you mentioned vomiting earlier, have you been able to keep fluids down today?

`Remembers 14 Sep vomiting`
`Rule: persistent vomiting, poor intake stay open`

17 SEP

YOU

the fever is back and I feel worse than before

AGENT

The fever returning and feeling worse is concerning, especially since you were vomiting earlier. Fevers can be a sign your body is fighting an infection. … Are you having any cold sweats or feeling like something terrible is about to happen?

`Source: MedlinePlus · Fever`
`Rule: fever → SEE A DOCTOR TODAY`

24 SEP

YOU

hi

AGENT

Hello. Last we spoke, you said you were feeling completely better with no fever and eating normally. Are you still feeling well, or is something bothering you now?

`Recalls the 20 Sep conversation`
`1 sentence shown as "Not from a cited source"`

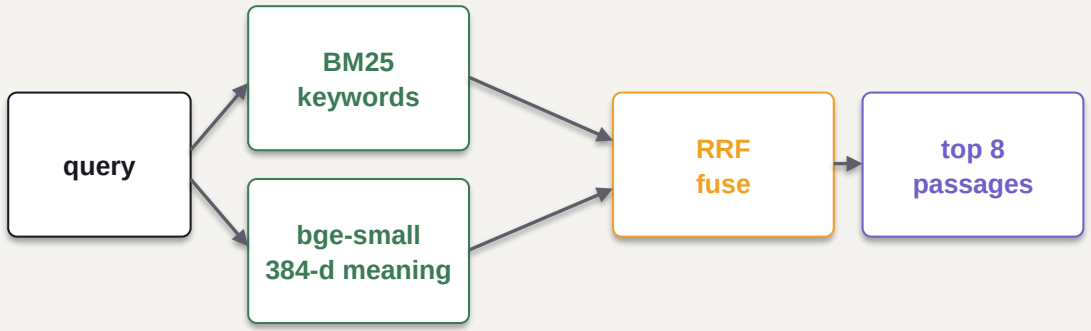
Per message: safety rules read it first → retrieval → AI reply (T = 0.3) with quoted spans → code checks spans → separate AI extractor (T = 0) → rule sets urgency, which only ever rises.

RAG: the agent may only quote from a local library

Retrieval-augmented generation: find relevant passages first, then let the model answer from them.

Layer	Content	Chunks from
1	consumer health text	MedlinePlus (NLM), 1,013 topics
2	how a night is measured	constants.py
3	when to seek care	red_flags.yaml
4	evidence for each cause	causes.yaml

1,849 chunks · ~400 tokens each · ~32 MB · no open-web search



FUSION

```
RRF(d) = Σ_lists 1 / (60 + rank(d))

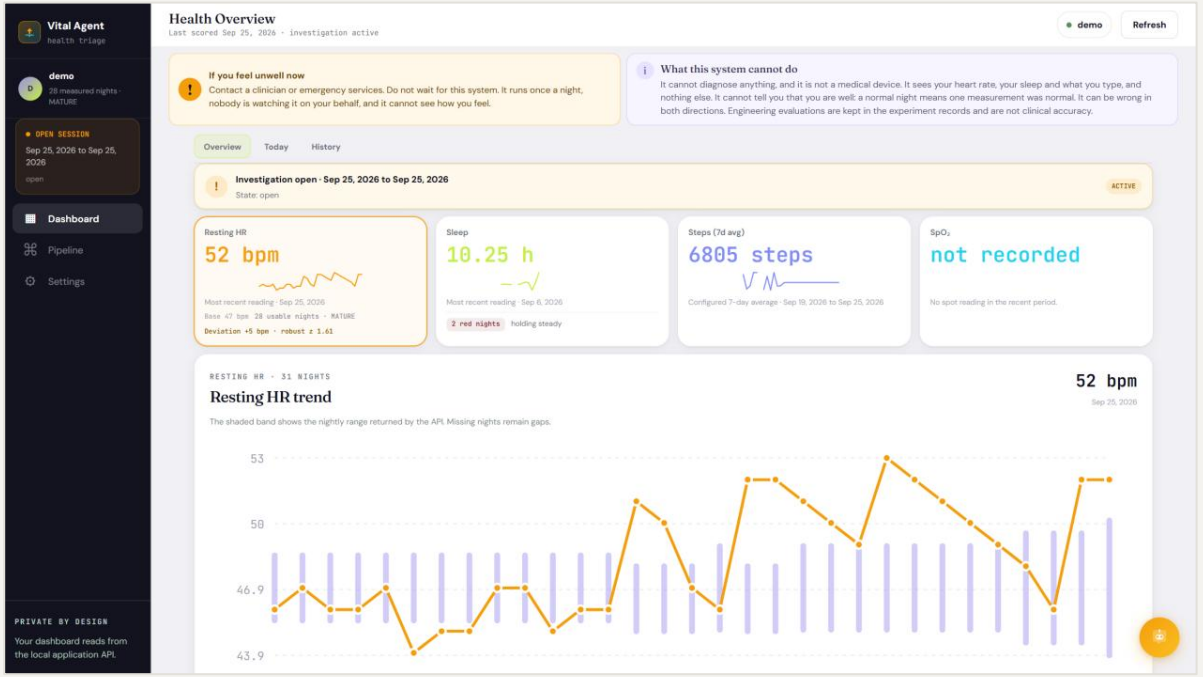
keep top 8, drop duplicates,
scan each for prompt injection
```

GROUNDING CHECK

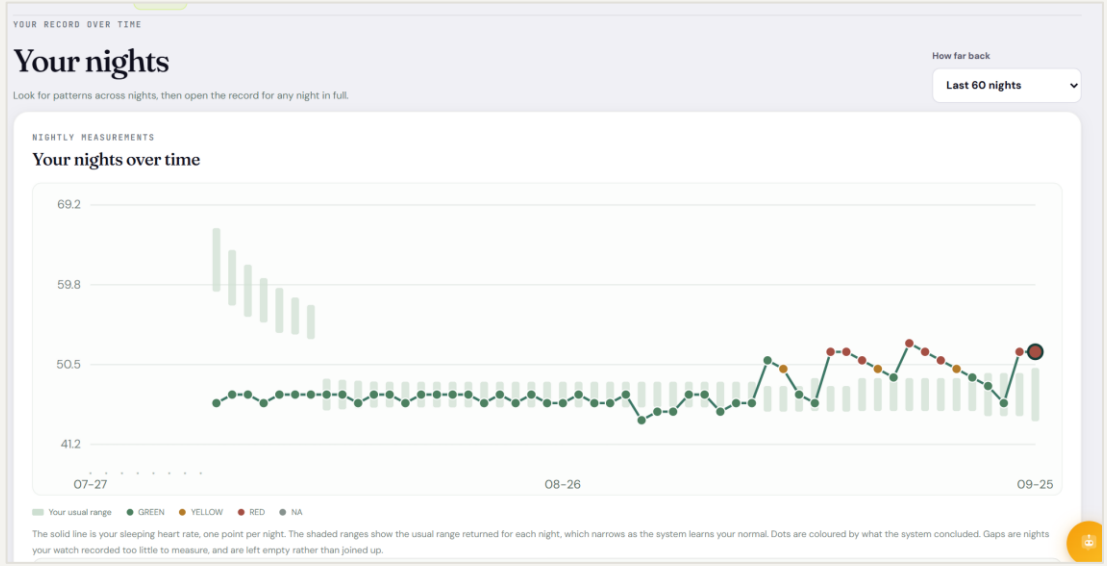
```
grounded ⇔ span ⊂ passage
   (exact text, case/space-folded)
no span or not found → claim stripped
citation = passage the code found it in
```

Dashboard: Overview, Today, History

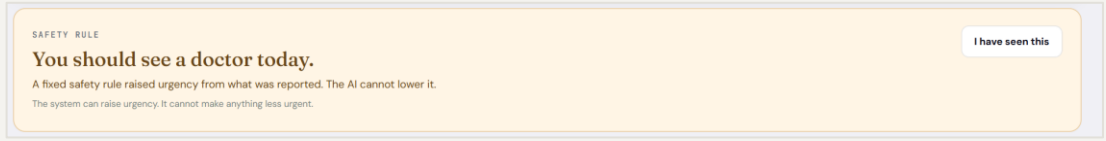
React + Vite, reading everything from the local API.



OVERVIEW · metric cards, baseline, z-score, resting-HR trend



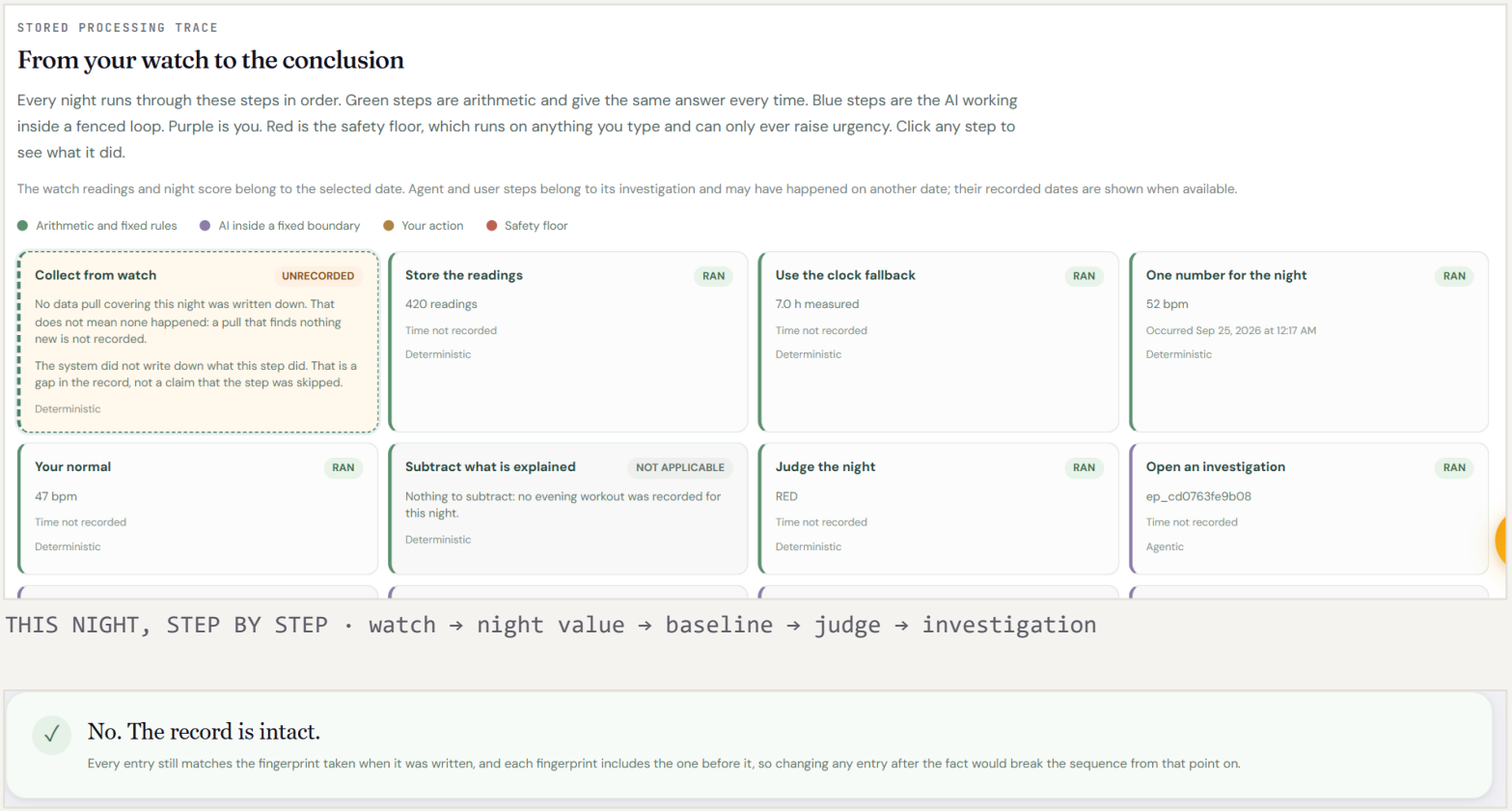
HISTORY · nights coloured GREEN / YELLOW / RED vs your usual range



Always on screen: "If you feel unwell now, contact a clinician" and "What this system cannot do". The chat button sits bottom-right on every page.

Pipeline: every step the system took, and proof it wasn't edited

The trace uses the same colours as this deck: green arithmetic, purple AI, amber you, red safety floor.



THIS NIGHT, STEP BY STEP · watch → night value → baseline → judge → investigation

✓ No. The record is intact.

Every entry still matches the fingerprint taken when it was written, and each fingerprint includes the one before it, so changing any entry after the fact would break the sequence from that point on.

Hash-chained audit log

hash_i = H(entry_i + hash_{i−1})
Change any entry and every later hash stops matching.

Four tabs

System overview: the three layers
This night, step by step: the trace
Everything the system did: audit + sentences
How this works: live provenance for every number

What is built, and what is still unproven

An inspectable design, not evidence that it helps anyone.

Built and running

- Watch → phone bridge → local service → dashboard + Telegram
- Personal baseline and a two-RED-night trigger, all arithmetic
- Bayesian belief over 12 causes and an EVSI question policy
- Bounded LangGraph agent with pause / resume across restarts
- Local RAG library with a code-level grounding check
- Deterministic safety floor the AI cannot lower
- Tamper-evident audit log and the new React dashboard

Not yet shown

- Never tested on a patient; no clinical trial or peer review
- Real-world detection accuracy is unknown
- On one Fitbit COVID archive: comparable to NightSignal, not better
- Most answer likelihoods are assumptions, and the file says so
- Grounding proves the quoted words exist, not that the sentence is fair
- Sleep apnoea is invisible by design
- Not a medical device. If you feel unwell, contact a clinician.

If someone feels unwell, contact a clinician or emergency services. Do not wait for the system. Use it to explain and evaluate the engineering, not to make a healthcare decision.